

Near-Endpoint Gaussian Small-Ball Exponents for Ellipsoidal Cylinders

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Status. Candidate partial result, likely valid. Conjecture 1 of Latała–Oleszkiewicz is proved for centered ellipsoids and ellipsoidal cylinders. The general symmetric-convex-body case remains open in this investigation.

1 Source conjecture

Latała and Oleszkiewicz [1] consider the standard Gaussian measure γ_n on $\mathbb{R}^n$ and the inradius

$$w(K) = \sup\{r > 0 : B_2^n(0, r) \subset K\}$$

of a symmetric convex set. Their Conjecture 1, on PDF page 2 of arXiv:math/0501268, is the following.

The following related conjecture seems reasonable.

Conjecture 1 *For any $\kappa \in (0, 1)$ there exist positive constants $C = C(\kappa)$ and $w_0 = w_0(\kappa)$ such that for any convex symmetric set K in $\mathbb{R}^n$ with $\gamma_n(K) \leq 1/2$ and $w(K) \geq w_0$ we have*

$$\forall_{0 \leq t \leq 1} \gamma_n(tK) \leq (Ct)^{\kappa w^2(K)} \gamma_n(K). \quad (4)$$

Thus, for every $\kappa \in (0, 1)$, the aim is to find constants $C(\kappa), w_0(\kappa) > 0$ such that

$$\gamma_n(tK) \leq (C(\kappa)t)^{\kappa w(K)^2} \gamma_n(K), \quad 0 \leq t \leq 1, \quad (1)$$

whenever $\gamma_n(K) \leq 1/2$ and $w(K) \geq w_0(\kappa)$. The source proves the coefficient $1/4$ in place of every $\kappa < 1$.

2 Proof intuition

For an ellipsoid, the Gaussian measure of tK is the lower tail at t^2 of a weighted chi-square variable $X = \sum_i \lambda_i g_i^2$. The inradius condition bounds every weight by w^{-2} . After enlarging the ellipsoid so that its Gaussian measure is exactly $1/2$, the number 1 is a median of X . Since no single weight can dominate, a one-sided variance estimate forces the total spectral mass $m = \sum_i \lambda_i$ to satisfy $m \geq 1 - O(w^{-1})$.

The exact product Laplace transform of X then produces a lower-tail power mw^2 . As the inradius grows this power approaches the conjectured endpoint w^2 , so every fixed coefficient $\kappa < 1$ follows. The argument also tolerates zero eigenvalues, hence unbounded ellipsoidal cylinders.

3 Statement of the partial result

Theorem 1. Fix $\kappa \in (0, 1)$ and set

$$\eta = \frac{1 + \kappa}{2}, \quad C_\kappa = \exp(1/\eta), \quad w_{0,\kappa} = \max \left\{ \frac{2\sqrt{2}}{1 - \kappa}, \sqrt{2 \log 2} \right\}. \quad (2)$$

Let

$$K = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^r \lambda_i x_i^2 \leq 1 \right\}, \quad 1 \leq r \leq n, \quad \lambda_i > 0, \quad (3)$$

after an orthogonal change of coordinates. If $\gamma_n(K) \leq 1/2$ and $w(K) \geq w_{0,\kappa}$, then

$$\gamma_n(tK) \leq (C_\kappa t)^{\kappa w(K)^2} \gamma_n(K) \quad (0 \leq t \leq 1).$$

In particular, Conjecture 1 holds for every centered ellipsoid and every ellipsoidal cylinder.

4 Half-measure reduction

We use the B-inequality of Cordero-Erausquin, Fradelizi, and Maurey [2]: for a symmetric convex set K , the function $a \mapsto \log \gamma_n(e^a K)$ is concave. The following consequence is also used explicitly in the source paper.

Lemma 1. If $\gamma_n(K) \leq 1/2$, choose $q \geq 1$ so that $\gamma_n(qK) = 1/2$. Then

$$\frac{\gamma_n(tK)}{\gamma_n(K)} \leq \frac{\gamma_n(tqK)}{\gamma_n(qK)} \quad (0 < t \leq 1). \quad (4)$$

Proof. Write $F(a) = \log \gamma_n(e^a K)$, $A = \log q \geq 0$, and $h = \log t \leq 0$. For concave F , the backward increment $F(x + h) - F(x)$ is nondecreasing in x . Hence $F(h) - F(0) \leq F(A + h) - F(A)$, which is exactly (4) after exponentiation. $\square$

It is therefore enough to prove the theorem when $\gamma_n(K) = 1/2$. Indeed, qK has inradius $W = qw(K) \geq w(K)$. When $C_\kappa t \leq 1$, the power with W^2 is no larger than the power with $w(K)^2$; when $C_\kappa t \geq 1$, the conclusion follows directly from $tK \subset K$.

5 The weighted chi-square estimate

Assume now that $\gamma_n(K) = 1/2$ and write $W = w(K)$. If $g_1, \dots, g_r$ are independent standard Gaussians, set

$$X = \sum_{i=1}^r \lambda_i g_i^2, \quad m = \mathbb{E}X = \sum_{i=1}^r \lambda_i, \quad v = \text{Var}(X) = 2 \sum_{i=1}^r \lambda_i^2.$$

Then

$$\gamma_n(tK) = \mathbb{P}\{X \leq t^2\}, \quad \mathbb{P}\{X \leq 1\} = \mathbb{P}\{X \geq 1\} = \frac{1}{2}. \quad (5)$$

The inradius identity is

$$\max_i \lambda_i = W^{-2}, \quad v \leq 2W^{-2}m. \quad (6)$$

Lemma 2 (Median-to-mean estimate). *Under (5)–(6),*

$$m \geq 1 - \frac{\sqrt{2}}{W}.$$

Proof. The assertion is immediate if $m \geq 1$. Suppose $m < 1$. Cantelli's one-sided variance inequality gives

$$\frac{1}{2} = \mathbb{P}\{X - m \geq 1 - m\} \leq \frac{v}{v + (1 - m)^2}.$$

Consequently $(1 - m)^2 \leq v \leq 2m/W^2 \leq 2/W^2$, proving the claim. For completeness, Cantelli's inequality follows by applying Markov's inequality to $(Y + v/a)^2$ when $\mathbb{E}Y = 0$, $\mathbb{E}Y^2 = v$, and $a > 0$. $\square$

Lemma 3 (Lower-tail estimate). *For every $t > 0$,*

$$\mathbb{P}\{X \leq t^2\} \leq e^{W^2/2} t^{mW^2}. \quad (7)$$

Proof. For $s > 0$, Markov's inequality and the Gaussian Laplace transform give

$$\mathbb{P}\{X \leq t^2\} \leq e^{st^2} \mathbb{E}e^{-sX} = e^{st^2} \prod_i (1 + 2s\lambda_i)^{-1/2}. \quad (8)$$

Put $L = W^{-2}$. The function $f(x) = \log(1 + 2sx)$ is concave and vanishes at zero, so $f(x) \geq (x/L)f(L)$ for $0 \leq x \leq L$. Therefore

$$\sum_i \log(1 + 2s\lambda_i) \geq mW^2 \log(1 + 2s/W^2).$$

Choosing $s = W^2/(2t^2)$ in (8) yields

$$\mathbb{P}\{X \leq t^2\} \leq e^{W^2/2} (1 + t^{-2})^{-mW^2/2} \leq e^{W^2/2} t^{mW^2}.$$

$\square$

6 Proof of Theorem 1

By Lemma 1, assume $\gamma_n(K) = 1/2$. The first lower bound in (2) and Lemma 2 imply

$$m \geq 1 - \frac{\sqrt{2}}{W} \geq \eta.$$

For $0 < t \leq C_\kappa^{-1}$, Lemma 3 gives

$$\begin{aligned} \frac{\gamma_n(tK)}{(C_\kappa t)^{\kappa W^2}/2} &\leq 2e^{W^2/2} C_\kappa^{-\kappa W^2} t^{(m-\kappa)W^2} \\ &\leq 2e^{W^2/2} C_\kappa^{-mW^2} \leq 2e^{-W^2/2} \leq 1. \end{aligned}$$

Here we used $\log C_\kappa = 1/\eta$, $m \geq \eta$, and $W \geq \sqrt{2 \log 2}$. If $C_\kappa^{-1} \leq t \leq 1$, then

$$\gamma_n(tK) \leq \gamma_n(K) = \frac{1}{2} \leq (C_\kappa t)^{\kappa W^2} \frac{1}{2}.$$

The case $t = 0$ is immediate. The reduction following Lemma 1 completes the proof for the original K .

7 Scope, sharpness, and novelty

For Euclidean balls of Gaussian measure $1/2$, the squared radius is asymptotic to n , while the probability at scale $t \downarrow 0$ has power n . Thus coefficient 1 is the natural endpoint and cannot be replaced by a larger universal coefficient. Theorem 1 reaches every strict sub-endpoint for quadratic bodies.

The factorized transform in (8) is the decisive special structure. For a general norm, the analogous random variable is a supremum of correlated Gaussian linear forms; the inradius bounds individual variances but does not supply the product estimate. Optimizing the source paper’s isoperimetric anchor is likewise capped at coefficient $1/4$. These two upgrade attempts did not yield the general conjecture.

A bounded novelty search on 11 August 2026 covered the run indexes, the exact arXiv id and title, exact-statement searches involving $\gamma_n(tK)$ and κw^2 , and combinations of “Gaussian small ball”, “inradius”, “ellipsoid”, and “weighted chi-square”. No explicit near-endpoint ellipsoidal theorem or full resolution of Conjecture 1 was found. This supports but does not certify novelty.

References

- [1] R. Latała and K. Oleszkiewicz, *Small ball probability estimates in terms of width*, *Studia Math.* **169** (2005), no. 3, 305–314; arXiv:math/0501268.
- [2] D. Cordero-Erausquin, M. Fradelizi, and B. Maurey, *The (B) conjecture for the Gaussian measure of dilates of symmetric convex sets and related problems*, *J. Funct. Anal.* **214** (2004), no. 2, 410–427.